Activity - 4.

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Hashing.

Given input {4371, 1323, 6173, 4199, 4344, 9679, 1989} hashing function $h(x) = x \pmod{10}$

Compute hash value.

Key	$h(x) = x \bmod 10$.
4371	1
1323	3
6173	3
4199	9
4344	4
9679	9
1189.	9.

a) Separate Chaining Hash Table:

Index	Chain.
0	-
1	4371
2	-
3	1323 -> 6173
4	4344
5	-
6	-
7	-
8	-
9	4199 -> 9679 -> 1989.

b) Open Addressing using Linear problem:

-> Insertions:

* 4371 -> 1
* 1323 -> 3
* 6173 -> 3 -> 4.
* 4344 -> 4 -> 5.
* 4199 -> 9.
* 9679 -> 9 -> 10.
* 1989 -> 9 -> 0 -> 1 -> 2.

Final Table.

Index	Value.
0	9679.
1	4371
2	1989.
3	1323
4	6173
5	4344
6	-
7	
8	-
9	4199.

c) open Addressing using Quadratic Problem

-> formula:

$$h_i(x) = (h(x) + i^2) \bmod 10.$$

Insertions.

4371 -> 1

1323 -> 3.

6173 -> 3 -> 4.

4344 -> 4 -> 5.

4199 -> 9.

9679 -> 9 -> 0.

1989 -> 9 -> 0 -> 3 -> 8.

Final table:

Index	Value:
0	9679.
1	4371
2	-
3	1323
4	6173
5	4344
6	-
7	-
8	1989
9	4199

d) Open Addressing Using Double Hashing.

Compute second hash.

Key	X Mod 7.	$h_2(k)$
4371	3	4
1322	0	7
6173	6	1
4199	6	1
4344	4	3
9679.	5	2
1989.	1	6.

Insertions:

* 4371 -> 1
* 1323 -> 3.
* 6173 -> 3 -> (3+1) = 4.
* 4344 -> 4 -> (4+3) = 7.
* 4199 -> 9.
* 9679 -> 9 -> (9+2) = 11.
* 1989 -> 9 -> (9+6) = 15.

Final table

Index	Value
0	-
1	4371
2	-
3	1326
4	6173
5	9673
6	-
7	4344
8	-
9.	4199.